DPP-ADAPT

A method for **A**rchitecture-**D**riven **A**doption of Digital Product Passport-enabled **P**rocurement **T**ransition

Master's thesis — Presentation and First Evaluation of a Design Science Research artifact
Public Sector Innovation and eGovernance M. Sc. (PIONEER)
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Expert Evaluation Questionnaire

https://forms.cloud.microsoft/e/uLjYfAX0wu

Expert Evaluation: Friendly Instructions

1. Check this method presentation and DPP-ADAPT
   Skim for relevance, completeness, consistency and fit to the problem dimensions.
2. Open the Excel workbook
   Look through P0 (readiness), P2 (attributes), P4 (patterns) especially for authority context.
3. Walk a use case through the views
   You can use the notebook example or select a suitable product from your context.
4. Think about a few criteria
   → Relevance, completeness, internal consistency, usability, fidelity, transferability.
5. Consider possible deal-breakers
   What would block adoption in your opinion? Where does the method underestimate complexity?

We will go through the presentation again during the evaluation interview.
In addition – or if you cannot find the time for an interview – please access the QR code/hyperlink on the first slide.

Thank you very much for your support.

Public procurement is being asked to use a data infrastructure it cannot read (yet).

Literature can show that three main factors actively influence the current challenge.

1

ESPR & DPP

The EU Ecodesign for Sustainable Products Regulation (ESPR 2024/1781) introduces Digital Product Passports as “regulated, machine-readable product data infrastructure”.

2

Strategic procurement

Public buyers are increasingly expected to drive sustainable procurement through specifications, qualification, award criteria, and contract management.

3

Integration gap

Authorities lack governance, semantic mapping, evidence logic, and integration patterns to translate DPP data into procurement-defensible decisions.

The major challenge is a missing decision architecture.

Interviews show that the challenge is not purely technical.

Public procurement organisations lack an **enterprise-architecture-based integration logic** for integrating externally governed DPP data into procurement-relevant, reliable, and legally defensible decisions.

Source: literature synthesis triangulated with 11 experts through video interviews.

Reframing

DPP-procurement integration is not currently an interface challenge but can be better understood as an EA problem.

→ Solving it requires governance, semantics mapping, technical pattern selection beyond connecting an API.

Why a method, not a reference architecture or a checklist.

Three artifacts were considered.

Reference architecture

Prescribes a target state that low-maturity authorities **cannot reach** without substantial preparation.

not preferred

Procedural checklist

Underestimates the architectural and governance dimensions; suitable only for narrow tasks.

not preferred

Method (DPP-ADAPT)

Combines architectural scope with applicable decision instruments/tools.
Anchored to TOGAF ADM.

meets expectations

Design Science Research + Method Engineering (Peffers et al., 2007; Brinkkemper, 1996; Henderson-Sellers & Ralyté, 2010)

DPP-ADAPT consists of two combined components + optional reference views.

Method and workbook are used together.

Method specification

This presentation + thesis document. Defines 7 phases (P0–P6), six method components per phase, decision criteria, validity rules.

Workbook

13 sheets in Excel: dashboard, instructions, ten phase sheets, reference data. Scoring + sign-off.

optional addition

Architectural views

Three views in ArchiMate and BPMN, each mapped to a specific phase.

There are seven phases designed around TOGAF's ADM, adapted for DPP-PP.

P0	P1	P2	P3	P4	P5	P6
Readiness & setup	Architecture vision	Business architecture	Information & application	Technology & integration pattern	Transition & operating model	Implementation governance & eval.
TOGAF Prelim	*TOGAF A*	*TOGAF B*	*TOGAF C*	*TOGAF D*	*TOGAF E + F*	*TOGAF G + H*

Each phase is specified along six method components:

inputs, activities, techniques, tools, roles and outputs including decision criteria and validity rules.

…and few architectual views to describe key phases.

R1 *Phase P0 + P3*

Overview Procurement process + DPP

BPMN

R2 *Phase P1*

Overview Application Layer

ArchiMate

R3 *Phase P2*

Manual Integration Pattern

Archimate

→ **Available views can be accessed here: DPP-ADAPT_arch_views**

Purpose: Assess readiness + set governance and decision rights.

Inputs

- Procurement & sustainability strategy
- Application inventory
- Sponsor candidates

Activities

- 25-statement readiness self-assessment
(5 dimensions × 5 statements)
- Stakeholder register & RACI
- Preliminary operating-model choice

Outputs

- Readiness level (Low/Medium/High)
- Approved RACI
- Sponsor sign-off

In the workbook: Sheets 3 (Readiness) and 4 (RACI). The overall score is auto-computed and traffic light colored (red/orange/green).

P1 · Architecture vision: motivation and scope

Purpose: Confirm motivation, choose product family, define in-scope process stages

Inputs

- P0 outputs
- Sustainability objectives
- Recurring procurements

Activities

- Drivers and goals
- ONE product family selection
- In-scope stage selection
- Capability gap

Outputs

- Scope canvas
- Capability gap analysis

Decision criterion: Exactly one product family in scope for first implementation. If governance dimension low, plan P6 in parallel with P5.

In the workbook: Sheet 5: Y/N drop-downs for stages, 1–5 scoring for drivers and capability gap.

Purpose: Select touchpoints, score DPP attributes, decide procurement-use roles.

Inputs

- P1 outputs
- DPP attribute candidates (illustrative)
- Procurement-law guidance

Activities

- Touchpoint matrix
- Attribute scoring on relevance (R), evidence confidence (E), and verifiability (V) on 1–5
- Procurement-use role decisions

Outputs

- Approved attributes by use-role
- View R1 (BPMN)
- Risk-gate outcome

In the workbook: Sheet 7: incl. each attribute's scale and corresponding item (EU energy label, ESPR, REACH, etc.).

Purpose: Semantic mapping, information lifecycle, application inventory

Inputs

- P2 outputs
- Application inventory
- Semantic references

Activities

- Per-attribute semantic mapping with type, rule, unit alignment, sign-off
- Information lifecycle (retention ≥60 mo)
- Apps in R2

Outputs

- Approved mapping table
- Information lifecycle
- View R2

In the workbook: Sheet 8: drop-downs for mapping type, status, unit alignment. Sheet 9 — applications and retention check.

Purpose: Score patterns A–E; document rationale.

Inputs

- P0 readiness
- P2 touchpoints
- P3 mapping

Activities

- Score each pattern on 7 dimensions
- Calculate weighted total
- Document reasoning statement

Outputs

- Chosen pattern (A/B/C)
- Weighted total scoring

In the workbook: Sheet 10: pattern scoring with weight-sum check. Result auto-pulled into the dashboard.

Purpose: Plateau roadmap, work packages, risks and risk management

Inputs

- P0–P4 outputs
- Portfolio context

Activities

- Three plateaus (current/first/scaled)
- Work-package register
- Risk register with RAG

Outputs

- Roadmap
- Work packages with dependencies
- Risk register

Decision criterion: Plateau 1 reachable within around 12 months by default.

In the workbook: Sheet 11: M = months from project start, PD = person-days, WP = work package. Drop-downs for WP status and plateau.

P6 · Implementation governance and evaluation

Purpose: Governance, indicators, expert review

Inputs

- P5 outputs
- Existing governance bodies

Activities

- Governance charter
- Indicator catalogue
- Ex-ante expert review

Outputs

- Complete Charter
- Indicators with targets
- Expert review report

Decision criterion: Panel ≥3 experts from PP/EA/sustainability/legal. Criteria avg. <3.0 triggers improvement before broader roll-out.

In the workbook: Sheet 12: expert ratings 1–5, indicators with target vs. current and last-review date.

Three integration patterns to choose from according to maturity level.

Each approach comes with a different level of integration across technology in use.

A	B	C
View R3		
Manual portal lookup	**Retrieval Service via Central Database**	**Retrieval Service via Data Space Environment**
Maturity fit: **Low**	Maturity fit: **Medium**	Maturity fit: **Medium–High**
Governance load: **Low**	Governance load: **Medium**	Governance load: **Medium–High**

Pattern selection is scored on 7 dimensions (Sheet 10): readiness fit, use-case fit, scalability, governance load, cost, time-to-value, future-proofness.

Default weights and rationale are on Sheet 2.

Worked example: laptop procurement

Fraunhofer ISST has been chose to demonstrate the method in a working environment.
→ See workbook for full example (Dashboard)

Fraunhofer ISST, headquartered in Dortmund, Germany, has approx. 120 notebook users on a 3 to 4 year refresh (€200–400K/cycle) running on SAP S/4HANA + SAP Ariba via central Fraunhofer ICT and Microsoft Entra ID, combined with in-house dataspace expertise and existing Kubernetes capability.

Fraunhofer ISST	
Staff	**191**
Operating budget	**€ 12.25 M / yr**
Notebook users	**approx. 120**
Refresh cycle	**3–4 years**
Cycle cost	**€ 200–400 K**
e-Procurement	**SAP Ariba**
ERP	**SAP S/4HANA**
IDP	**MS Entra ID**
Directive	**2014/24/EU**

Readiness 3.88 → HIGH

Chosen
Pattern C

Outcome: Pattern C selected (procurement-platform retrieval service via Data Spaces).
Two criteria approved (energy efficiency, repairability). 12-month plateau plan to first DPP-informed tender.

Workbook contains 12 sheets, every output is a single checkpoint.

1	**Dashboard**	Steering-committee view + charts
2	**Method map**	Weights and rationale, version log
3	**P0 Readiness**	25 statements × 5 dimensions
4	**P0 RACI**	Stakeholders + decision rights
5	**P1 Scope**	Drivers, goals, product family
6	**P2 Touchpoints**	Stage × use-role matrix
7	**P2 Attributes**	Scale, meaning, source, score
8	**P3 Mapping**	Semantic mapping & sign-off
9	**P3 Apps**	Application & info lifecycle
10	**P4 Pattern**	7-dimension scoring & choice
11	**P5 Roadmap**	Plateaus, WPs (M, PD), risks
12	**P6 Evaluation**	Indicators & expert ratings

Thank you for your time. Please fill in the questionnaire.

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